Lecture 20

Fibre bundles

Objectives. *A vector field assigns a vector in a different tangent space at each point. A bundle describes these fibres and the changes between their local coordinates. We reconstruct bundles from transition functions and prove the global-section criterion for principal bundles. The Hopf bundle and the tangent sphere show how local products can fail to give a global product.*

By the end of this lecture you will be able to:

- Construct a smooth vector or principal bundle from a smooth cocycle.
- Relate local frames, section components and transition functions.
- Prove the principal-bundle section criterion and identify the Hopf bundle.
- Compute the tangent sphere's transition winding and compare it with Hopf's.

Unit 20A Bundles and gluing

20.1 Local products and fibres

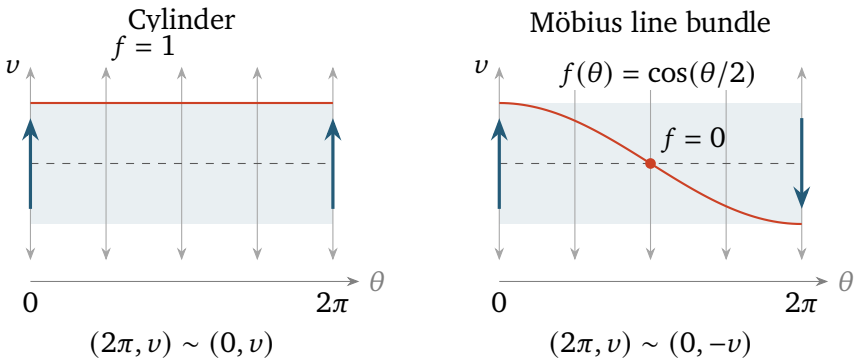
Example 20.1 (Cylinder and Möbius gluing)**.** We attach a real line to each point of a circle. The product $S^1 \times \mathbb{R}$ gives a cylinder. A different gluing gives the Möbius line bundle:

$$E_{\mathrm{M}} = ([0, 2\pi] \times \mathbb{R})/\!\sim, \qquad (2\pi, v) \sim (0, -v).$$

Over a small arc of the circle, both spaces are products with $\mathbb{R}$. A full circuit reverses the chosen fibre coordinate in E_{M}. The familiar finite strip contains only part of each full line fibre.

The cylinder has a continuous choice of a nonzero vector with constant coefficient 1. A continuous choice of a vector in each fibre of E_{M} has a coefficient f on $[0, 2\pi]$ with $f(2\pi) = -f(0)$. The intermediate value theorem forces a zero. Thus these local products have different global properties.

The cut-open products show the gluing and the full line fibres.



Shading: $|v| \le 1$; arrows continue along the full fibre $\mathbb{R}$.

Figure 20.1: The cylinder identifies equal fibre coordinates at its ends. The Möbius line bundle identifies opposite coordinates; shading marks only the strip $|v| \le 1$ in each full line fibre.

Definition 20.2 (Fibre bundle)**.** A smooth *fibre bundle* consists of a smooth surjection $\pi_E : E \to M$ and a manifold F, its *typical fibre*. Every point of the *base space* M has an open neighbourhood U_i and a diffeomorphism

$$\varphi_i : U_i \times F \longrightarrow \pi_E^{-1}(U_i), \qquad \pi_E(\varphi_i(x, v)) = x.$$

This is a *local trivialization*. The manifold E is the *total space*. The *fibre at* x is $E_x = \pi_E^{-1}(x)$.

All manifolds here are Hausdorff and second countable.

Definition 20.3 (Bundle isomorphism and triviality)**.** A *bundle isomorphism* over M is a diffeomorphism of total spaces commuting with the projections and preserving the specified fibre structure. A bundle is *trivial* if it is isomorphic to $M \times F$.

A local trivialization assigns a fibre coordinate independently at each base point of U_i.

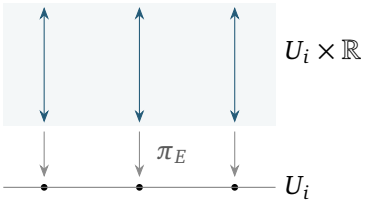


Figure 20.2: A local trivialization gives fibre coordinates over U_i. The projection sends each vertical fibre to its base point.

Definition 20.4 (Vector bundle)**.** A real or complex *vector bundle of rank* r has vector-space fibres and local trivializations linear on each fibre, with typical fibre $V_{\text{rep}} = \mathbb{R}^r$ or $\mathbb{C}^r$.

Definition 20.5 (Line bundle)**.** A rank-one vector bundle is a *line bundle*.

For example, TM of Lecture 8 is a rank-dim M real vector bundle.

Definition 20.6 (Section)**.** A *section over an open set* $U \subset M$ is a smooth map $s : U \to E$ such that $\pi_E \circ s = \mathrm{id}_U$. It is a *global section* when $U = M$.

Definition 20.7 (Local frame). A *local frame* of a rank-r vector bundle is a row of r smooth sections that form a basis in each fibre of its domain.

The frames of a single vector space have no preferred member. Given two row frames p and p', exactly one invertible matrix a satisfies $p' = pa$. Collecting the frames of all fibres gives a space over M. Its points are bases. Right multiplication changes those bases. A principal bundle records this space of choices and the group relating them.

Definition 20.8 (Principal bundle). Let G be a Lie group. A *principal G-bundle* is a bundle $\pi_E : P \to M$ with a smooth right action of G that is free and transitive on each fibre. Its local trivializations are required to satisfy $\varphi_i(x, ab) = \varphi_i(x, a)b$.

Freeness and transitivity say that any two points of a fibre differ by exactly one group element. Choosing $\varphi_i(x, e)$ chooses a reference in that fibre. The group identity belongs to this coordinate description. It is not a preferred point of the fibre. For the frame bundle, the reference is a basis. For the Hopf bundle, it will be a unit representative of a complex line.

Lecture 4's smooth covering examples are bundles with discrete fibre. The sheets over an evenly covered set give the local product. The antipodal cover $S^n \to \mathbb{R}\mathrm{P}^n$ is a principal $\mathbb{Z}_2$-bundle under the action $x \mapsto \pm x$. The rotation cover $S^3 \to \mathrm{SO}(3)$ has the same property. A general covering need not have a deck group transitive on its fibres.

20.2 Transition functions and reconstruction

Definition 20.9 (Transition functions and structure group). For a vector bundle, comparing trivializations defines smooth *transition functions* $g_{ij} : U_i \cap U_j \to \mathrm{GL}(r, \mathbb{K})$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, by

$$\varphi_j(x, v) = \varphi_i(x, g_{ij}(x)v).$$

Thus the coordinates of the same vector satisfy $v_i = g_{ij}v_j$. The *structure group* is the group of allowed transition matrices, usually $\mathrm{GL}(r, \mathbb{R})$ or $\mathrm{GL}(r, \mathbb{C})$.

The symbols g_{ij} denote transitions, and h_i will denote frame changes. Restricting to oriented orthonormal frames makes the structure group $\mathrm{SO}(r)$. For a principal bundle, equivariance instead gives

$$\varphi_j(x, a) = \varphi_i(x, g_{ij}(x)a), \qquad g_{ij} : U_i \cap U_j \to G.$$

Indeed, compare the trivializations at the group identity and then act on the right by a. This also shows that the principal transitions are smooth. For a vector bundle, the standard bases give local row frames e_i with $e_j = e_i g_{ij}$. On a triple overlap, successive coordinate changes give

$$v_i = g_{ij}v_j = g_{ij}g_{jk}v_k.$$

Direct comparison gives $v_i = g_{ik}v_k$ for every $v_k \in V_{\mathrm{rep}}$. Thus composition forces the product law below.

Definition 20.10 (Cocycle). The *cocycle condition* is

$$g_{ij}g_{jk} = g_{ik}, \qquad g_{ii} = I, \qquad g_{ji} = g_{ij}^{-1}.$$

For a principal bundle, replace the identity matrix I by the group identity e.

The coordinate arrows go from k through j to i, in the order of matrix action on a column.

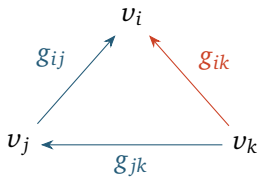


Figure 20.3: A fibre-coordinate column passes from region k to j and then to i by $g_{ij}g_{jk}$. Direct passage uses g_{ik}. All maps refer to the same base point in a triple overlap.

We describe the Möbius gluing by two base arcs. Let U_1 and U_2 be the images of the angle intervals $(-\pi/2, 3\pi/2)$ and $(\pi/2, 5\pi/2)$. Their overlap has a component A containing angle π and a component B containing angle 0. On A the angle representatives agree. On B they differ by 2π, so the Möbius seam reverses the coefficient. The transition is therefore

$$g_{12}|_A = +1, \qquad g_{12}|_B = -1.$$

For the cylinder both signs are $+1$. The picture records these overlaps and the effect of changing a frame on one whole arc.

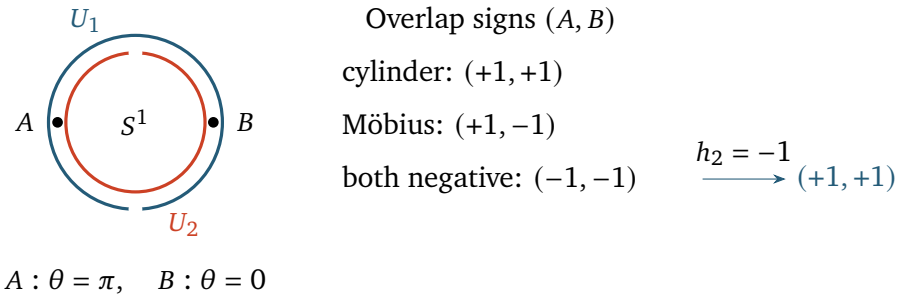


Figure 20.4: The arcs overlap near π in A and near 0 in B. Changing one arc's frame by -1 reverses both overlap signs.

Proposition 20.11 (Cocycle reconstruction · *proved*)**.** *Smooth transition matrices satisfying the cocycle condition on an open cover of M determine a smooth vector bundle, up to bundle isomorphism. The same construction with fibre a Lie group G and transition functions acting by left multiplication constructs a principal bundle.*

Proof. We glue product charts and verify that their quotient is a smooth manifold. Take the disjoint union of $U_i \times V_{\mathrm{rep}}$ and identify

$$(x, v)_j \sim (x, g_{ij}(x)v)_i.$$

The identity law gives reflexivity. The inverse law gives symmetry. The product law gives transitivity, so this is an equivalence relation. Project its quotient to M by the common base point. Over U_i, each class has a unique representative in the ith copy. The corresponding local coordinate map sends a representative in copy j to $(x, g_{ij}(x)v)$. It is a homeomorphism because the preimage of an open set in one copy is open in every other copy, since the transition map is a diffeomorphism. The quotient topology therefore agrees with these product charts. Their changes are smooth and linear on fibres.

Points with different base points have disjoint neighbourhoods lifted from M. Points with the same base point can be separated in one product chart. Thus the total space is Hausdorff. Second countability of M gives a countable subcover. Choose a basis element inside a cover member at each point and choose one member per basis element. The corresponding product charts, each second countable, give a countable basis for the total space. Hence the quotient is a smooth manifold.

The same argument with V_{rep} replaced by G gives smooth charts $U_i \times G$. The action $(x, a) \cdot b = (x, ab)$ is independent of the chart because left transition multiplication commutes with right multiplication. It is free and transitive on each fibre, as required in the principal definition.

If another bundle has trivializations φ_i with these transitions, send the quotient class $[(x, v)_i]$ to $\varphi_i(x, v)$. The defining transition relation makes this independent of its representative. In product charts it is the identity, so it is a smooth bundle isomorphism, equivariant in the principal case. This proves uniqueness. ∎

For the two-arc data $(+1, -1)$, the quotient is E_{M}. The product charts use exactly its angle representatives and seam relation.

Proposition 20.12 (Change of frame · *proved*). *Let $h_i \colon U_i \to \mathrm{GL}(r, \mathbb{K})$ be smooth frame changes. Under changes $e'_i = e_i h_i$, the transitions become*

$$g'_{ij} = h_i^{-1} g_{ij} h_j.$$

For a principal bundle, smooth maps $h_i \colon U_i \to G$ change the local sections by $s'_i = s_i h_i$ and give the same law. Conversely, transitions related in this way give isomorphic bundles.

Proof. The new frames satisfy $e'_j = e'_i h_i^{-1} g_{ij} h_j$, proving the displayed change law. For an isomorphism between the two constructions, use $(x, v)_i \mapsto (x, h_i^{-1} v)'_i$. The change law makes this map agree on overlaps. Its inverse uses h_i, so it is a smooth bundle isomorphism. For principal sections the computation is $s'_j = s_i g_{ij} h_j = s'_i h_i^{-1} g_{ij} h_j$. The same chartwise map, with v a group element, is equivariant. ∎

Proposition 20.13 (Common refinement · *proved*). *Two covers can be compared by restricting their trivializations to the pairwise intersections of their members.*

Proof. The pairwise intersections form an open cover of M refining both covers. Restrictions preserve the transition relations. The chartwise isomorphism in Proposition 20.11 compares each restricted reconstruction with its original bundle. ∎

20.3 Sections and frames of a vector bundle

In a local frame, write $s = e_i \sigma_i$. Its coefficient columns satisfy

$$\sigma_i = g_{ij} \sigma_j.$$

Conversely, smooth columns with this compatibility glue to a section. The zero columns always give a global zero section.

A global frame exists exactly when the vector bundle is trivial. Indeed, a frame defines the isomorphism $(x, v) \mapsto e(x)v$. Its inverse is smooth in local coordinates because the frame matrix is invertible. A trivialization supplies the converse frame by its standard basis. For

a line bundle, one nowhere-zero section is a frame. For higher rank, one section need not give an entire basis.

In the Möbius line bundle, the intermediate value argument excludes a nowhere-zero section, so no global frame exists. The zero section gives no basis of a fibre.

Self-check. Describe the Möbius line bundle by its seam and its two overlap signs. Show that it is nontrivial. Determine what happens if both overlap signs are -1.

Answer. The seam is $(2\pi, v) \sim (0, -v)$, with transition signs $(+1, -1)$. A nowhere-zero section would have a continuous coefficient satisfying $f(2\pi) = -f(0)$, which the intermediate value theorem excludes. A trivial line bundle has a nowhere-zero section, so the bundle is nontrivial. Its zero section still exists. For signs $(-1, -1)$, change the frame on U_2 by -1. The change law gives $1^{-1}(-1)(-1) = +1$ on both components. These are the cylinder transitions, so this bundle is trivial.

Self-check. State in one sentence what a fibre bundle is and why we introduced it when studying vector fields. Explain what local product charts do not determine by themselves.

Answer. A fibre bundle is a smooth space over a base that is locally a product with a fixed fibre, introduced to describe the varying spaces in which fields take values. The overlap identifications determine how the products fit together. The cylinder and the Möbius line bundle have the same typical fibre $\mathbb{R}$ but different overlap signs, so local product shape alone does not determine triviality.

Unit 20B Principal bundles, Hopf and the tangent sphere

20.4 Principal and associated bundles

> **Definition 20.14** (Oriented orthonormal frame bundle)**.** For an oriented Riemannian n-manifold, its *oriented orthonormal frame bundle* consists of all positive orthonormal frames in its tangent spaces.

A point of this total space is a base point together with an ordered positive orthonormal basis of its tangent space. Right multiplication of a row frame by $\mathrm{SO}(n)$ is free and transitive on each fibre. Local Gram–Schmidt frames give equivariant product charts. This makes it a principal $\mathrm{SO}(n)$-bundle, even when no global tangent frame exists.

The local sections $s_i(x) = \varphi_i(x, e)$ obey $s_j = s_i g_{ij}$. The group acts on the right, while g_{ij} changes fibre coordinates by multiplication on the left.

> **Proposition 20.15** (The section criterion · *proved*)**.** *A principal bundle is trivial if and only if it has a global section.*

Proof. A global section s gives a map $\Psi\colon M \times G \to P$ by $\Psi(x, a) = s(x)a$. Freeness and transitivity make it bijective and equivariant. In a local trivialization write $s(x) = s_i(x)b_i(x)$. Then $\Psi(x, a)$ has local coordinate $b_i(x)a$. Its inverse has coordinate $b_i(x)^{-1}a$, so both maps are smooth. Conversely, a trivialization gives the section $x \mapsto (x, e)$. ∎

Vector-bundle triviality requires a full frame, as above.

Definition 20.16 (Associated bundle)**.** Let $\rho\colon G \to \mathrm{GL}(V_{\mathrm{rep}})$ be a smooth representation. The *associated bundle* is

$$P \times_G V_{\mathrm{rep}} = (P \times V_{\mathrm{rep}})/\!\sim, \qquad (pa, v) \sim (p, \rho(a)v), \qquad [p, v] \longmapsto \pi_E(p).$$

For the oriented orthonormal frame bundle, take the standard representation of $\mathrm{SO}(n)$ on $\mathbb{R}^n$. A row frame p and a component column v specify the tangent vector pv. The quotient recovers the tangent bundle by the map

$$P \times_{\mathrm{SO}(n)} \mathbb{R}^n \longrightarrow TM, \qquad [p, v] \longmapsto pv.$$

The equality $(pa)v = p(av)$ explains the identification in the quotient. Replacing p by pa changes the same vector's components from v to $a^{-1}v$. Thus $[pa, a^{-1}v] = [p, v]$. In a local frame the displayed map and its inverse are product-coordinate identity maps. Hence it is a smooth vector-bundle isomorphism.

For a general representation, the local trivializations are $(x, v) \mapsto [s_i(x), v]$. Their transitions are $\rho(g_{ij})$, so reconstruction gives a smooth vector bundle.

Definition 20.17 (Equivariant map)**.** A smooth map $f\colon P \to V_{\mathrm{rep}}$ is *equivariant* for this right action if

$$f(pa) = \rho(a)^{-1} f(p).$$

Sections correspond to these equivariant maps. The section is $x \mapsto [p, f(p)]$ for any p over x: $[pa, f(pa)] = [p, \rho(a) f(pa)] = [p, f(p)]$. Conversely, a section uniquely determines these fibre coordinates $f(p)$. In a product chart they are $\rho(a)^{-1} f(s_i(x))$, which proves smoothness and the converse.

20.5 Homogeneous spaces as principal bundles

Proposition 20.18 (Homogeneous principal bundles · *proved given the cited homogeneous-space and inverse function theorems*)**.** *For a closed subgroup H of a Lie group G, the quotient $G \to G/H$ is a principal H-bundle under right multiplication.*

Proof. We construct local sections of the quotient submersion and use the group action to trivialize its fibres. For a closed subgroup H of a Lie group G, Theorem 8.25 gives the smooth quotient G/H and a submersion $\pi_E\colon G \to G/H$. The inverse function theorem gives local submersion coordinates. Choose base coordinates y and supplement the independent differentials of $y \circ \pi_E$ by complementary coordinates ζ. The map $(y \circ \pi_E, \zeta)$ has invertible differential, hence is a local coordinate system. In these coordinates $\pi_E(y, \zeta) = y$. Fixing ζ gives a smooth local section s near each base point. The maps

$$U \times H \longrightarrow \pi_E^{-1}(U), \qquad (x, h) \longmapsto s(x)h$$

are diffeomorphisms. Indeed, their inverse sends a to $(aH, s(aH)^{-1}a)$. The second entry lies in the embedded subgroup H and is smooth there. In submanifold charts its normal coordinates vanish. Thus $H \to G \to G/H$ is a principal H-bundle. ∎

For example, the quotient $\mathrm{SO}(3) \to S^2$, given by the last column, has fibre $\mathrm{SO}(2)$. The first two columns give a positive orthonormal tangent frame at that point. This identifies $\mathrm{SO}(3)$ with the oriented orthonormal frame bundle of the round sphere.

20.6 The Hopf bundle

Physical input. A pure qubit state is a complex line in $\mathbb{C}^2$. Its normalized vector representatives differ by a common phase in $\mathrm{U}(1)$.

A phase convention chooses one unit representative of each line in its domain. The real tautological line gives a simpler example of choosing vectors on lines. Over $\mathbb{R}\mathrm{P}^1$, its fibre at a line is that line itself. Write its unit vector as $q(\alpha) = (\cos\alpha, \sin\alpha)^T$ for $0 \le \alpha \le \pi$. The lines at 0 and π agree. Their unit representatives satisfy $q(\pi) = -q(0)$. Thus the coefficient in $vq(\alpha)$ changes sign at the seam. This is the Möbius line bundle over $\mathbb{R}\mathrm{P}^1 \cong S^1$. The figure shows the lines and their pairs of unit vectors.

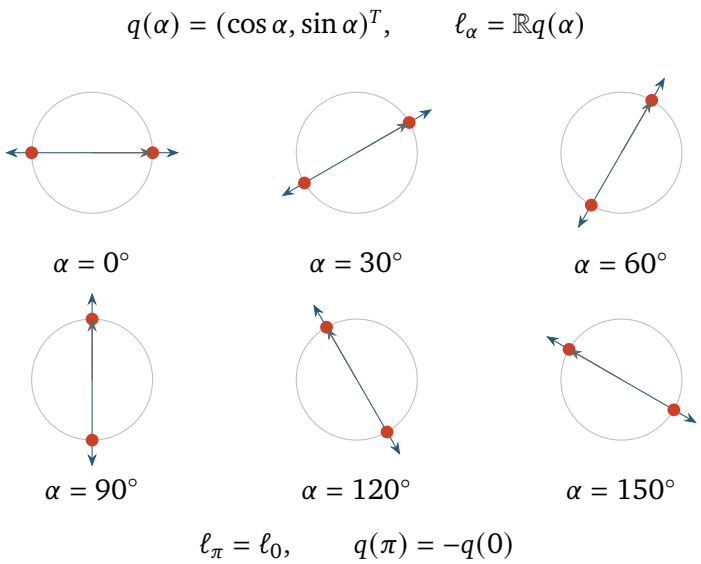


Figure 20.5: Each base point is a real line, and its fibre is that same line. The unit representatives change from $q(0)$ to $-q(0)$ as the line makes one circuit of $\mathbb{R}\mathrm{P}^1$.

Let $N = [1 : 0]$ and $S = [0 : 1]$ be the north and south poles of the Bloch sphere. The projective coordinate $z = b/a$ is defined on $U_N = \mathbb{C}\mathrm{P}^1 \setminus \{S\}$. The coordinate $w = a/b = 1/z$ is defined on $U_S = \mathbb{C}\mathrm{P}^1 \setminus \{N\}$. Unit representatives of their lines are

$$p_N(z) = \frac{(1, z)^T}{\sqrt{1 + |z|^2}}, \qquad p_S(w) = \frac{(w, 1)^T}{\sqrt{1 + |w|^2}}.$$

The northern representative tends to $(0, e^{i\phi})^T$ along $z = Re^{i\phi}$ as $R \to \infty$. Its limit at the south pole depends on the approach direction, so this phase convention does not extend there.

Definition 20.19 (Tautological line bundle)**.** The *tautological line bundle* is the set

$$\mathcal{T} = \{(\ell, v) \in \mathbb{C}\mathrm{P}^1 \times \mathbb{C}^2 : v \in \ell\}, \qquad (\ell, v) \longmapsto \ell.$$

The fibre at ℓ is the line ℓ itself, which explains the name. Its local coordinates are $v = cp_N(z)$ or $v = cp_S(w)$, with $c \in \mathbb{C}$. The inverse coefficient is $c = p_N^\dagger v$ or $c = p_S^\dagger v$, respectively. These maps give smooth product charts with linear overlap maps. On the overlap, $p_S = p_N|z|/z$. Orient the equator $|z| = 1$ by increasing ϕ, as the boundary of the northern cap. On this positive equator $z = e^{i\phi}$, the frame transition is $g_{NS} = e^{-i\phi}$, with winding -1. We keep the chart order (N, S). Reversing that order replaces the transition by its inverse and reverses its winding. For one fibre point, the phase coordinates obey

$$p_N e^{i\chi_N} = p_S e^{i\chi_S}, \qquad \chi_S = \chi_N + \phi \pmod{2\pi}.$$

The cap products glue by this change from northern to southern phase coordinates.

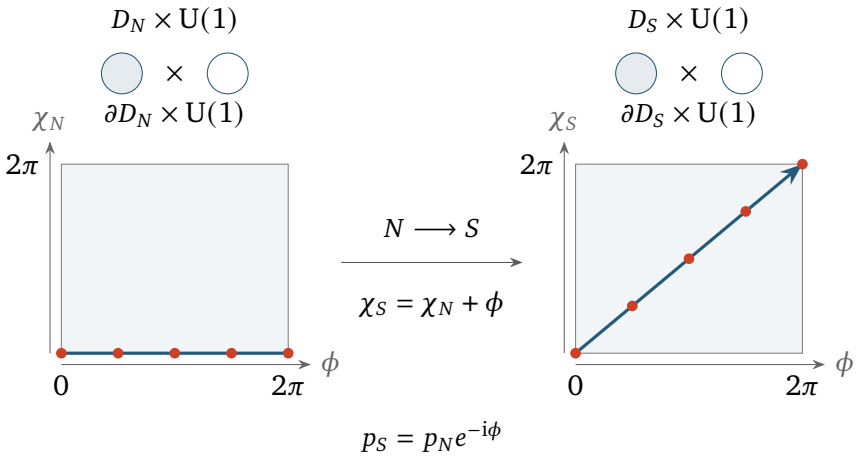


Figure 20.6: Each cap product has a boundary torus, shown cut open into a square. The same fibre point changes coordinates by $\chi_S = \chi_N + \phi$ from north to south, although the frame transition is $e^{-\mathrm{i}\phi}$.

The nontriviality proof compares the fundamental groups of a product and S^3. We complete Lecture 4's argument for Proposition 4.9, including the omitted-point step.

Lemma 20.20 (Sphere loops · *proved*)**.** *The spheres S^2 and S^3 are simply connected.*

Proof. To prove that S^n is simply connected for $n \in \{2, 3\}$, subdivide a loop into hemispherical pieces using Lemma 3.5. On each piece replace it by the normalized straight segment between its endpoints. Both curves have positive scalar product with the hemisphere centre, so their normalized straight interpolation is a homotopy fixing those endpoints. The resulting polygon lies in finitely many two-dimensional linear subspaces. Their union misses a point of the sphere. For each subspace choose a nonzero linear functional vanishing on it and evaluate these on $(1, t, \dots, t^n)$. Each gives a nonzero polynomial, so a parameter outside their finite root sets gives a vector outside the union. Normalize it to obtain the omitted sphere point. Stereographic projection from that point puts the polygon in $\mathbb{R}^n$, where straight contraction to its base point fixes that base point. Pasting the piece homotopies proves the assertion for $n = 2$ and $n = 3$. Both spheres are path-connected by short arcs, joining antipodes via a third point. ∎

Theorem 20.21 (The Hopf bundle · *proved given the cited homogeneous-space and inverse function theorems*)**.** *The map $\mathrm{SU}(2) \to \mathrm{SU}(2)/\mathrm{U}(1) \cong \mathbb{C}\mathrm{P}^1 \cong S^2$ is a nontrivial principal $\mathrm{U}(1)$-bundle. Under $\mathrm{SU}(2) \cong S^3$, it sends a unit column $(a, b)^T$ to its line $[a : b]$. Its associated complex line bundle for the standard $\mathrm{U}(1)$ representation is the tautological line bundle.*

Proof. Recall Lecture 8's explicit identification

$$(a, b) \longmapsto \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Right multiplication by $\mathrm{diag}(u, u^{-1})$, for $u \in \mathrm{U}(1)$, multiplies the first column by u. Its orbits are exactly the unit representatives of each complex line. The homogeneous-bundle proposition proves the principal structure.

By Lemma 20.20, both S^2 and S^3 are simply connected. Loops and based homotopies in a product are precisely pairs of loops and homotopies. Thus $\pi_1(S^2 \times S^1) \cong \mathbb{Z}$, whereas $\pi_1(S^3) = 0$. A trivial Hopf bundle would give a diffeomorphism between these total spaces, which is impossible.

The map $[p, v] \mapsto ([p], vp)$ identifies the associated line with $\mathcal{T}$. It is well-defined because $(pu, v) \sim (p, uv)$. Its inverse exists on each fibre, and in the local unit representatives above both maps are product-coordinate identity maps. Hence it is a smooth bundle isomorphism. ∎

By the section criterion, Hopf nontriviality excludes a global smooth phase convention. A continuous global section would also give a homeomorphism with $S^2 \times S^1$: in local coordinates the inverse uses multiplication by the inverse of the section's continuous phase. The same fundamental-group contradiction therefore excludes a continuous global choice. The local choices p_N and p_S represent the same state through their phase transition. Lecture 24 studies transport of these phases. The phase circle below lies over a single fixed state line.

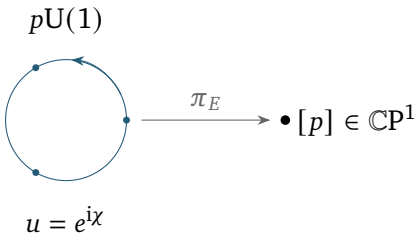


Figure 20.7: All unit vectors pu, with $u \in \mathrm{U}(1)$, lie over the same line $[p]$. The circle records their relative phase.

We recall from Lecture 3 that winding classifies loops in $\mathrm{U}(1)$ up to homotopy. For the bundle classification, we cite that bundles over either disk are trivial. Trivializing over the two caps leaves one transition on their overlap, which retracts onto the equator. Changing the section on a cap multiplies this transition by a phase that extends across a disk. Such a boundary phase has winding zero, since its loop contracts across that disk. Hence changing cap sections preserves the transition winding. The cited clutching theorem also says that homotopic equatorial transitions give isomorphic topological bundles. Together these facts explain why a single integer classifies them.

Proposition 20.22 (Clutching classification · *cited*)**.** *The underlying topological principal* $\mathrm{U}(1)$*-bundles over* S^2*, up to continuous equivariant bundle isomorphism, are classified by the winding of the transition* g_{NS} *on the positive equator. Every integer occurs.*

This is Naber, Theorem 4.4.3, pp. 231–232. That theorem concerns continuous bundles. No smooth classification proof is claimed here. Its integer invariant already distinguishes the smooth examples. On the overlap $z \neq 0$, the transitions $(z/|z|)^n$ give smooth representatives for every $n \in \mathbb{Z}$. The Hopf transition has winding -1, which gives a second proof of its nontriviality.

20.7 The tangent sphere and its transition winding

Use the oriented complex charts $z = x + \mathrm{i}y$ and $w = u + \mathrm{i}v = 1/z$ of Lecture 7 on $S^2 \cong \mathbb{C}\mathrm{P}^1$. Here u and v denote real chart coordinates. The symbol u in the Hopf action denoted a

phase, and v in the bundle charts denoted a fibre coordinate. Multiplication by i on tangent vectors is defined by $\mathrm{i}\partial_x = \partial_y$. The change $w = 1/z$ has complex-linear differential and preserves this rule. The real tangent planes therefore form a complex line bundle.

Example 20.23 (The tangent-sphere transition)**.** Find its frame transition and its component transition around the equator. Use complex frames $e_z = \partial_x$ and $e_w = \partial_u$, where a and b now denote real component coefficients. A complex coefficient $a + \mathrm{i}b$ means the real vector $a\partial_x + b\partial_y$ in the first frame. The coordinate derivative and its inverse are

$$\frac{\mathrm{d}w}{\mathrm{d}z} = -z^{-2}, \qquad \frac{\mathrm{d}z}{\mathrm{d}w} = -z^2.$$

The chain rule gives the frame relation and the component relation

$$e_w = e_z(-z^2), \qquad \sigma_w = (-z^{-2})\sigma_z.$$

The real plane rotation through β is

$$\mathrm{Rot}(\beta) = \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix}.$$

Identify a complex section coefficient with the real column $(\mathrm{Re}\,\sigma, \mathrm{Im}\,\sigma)^T$. On $z = e^{\mathrm{i}\phi}$ the corresponding real matrices are

$$g_{zw} = -\mathrm{Rot}(2\phi), \qquad \sigma_w = -\mathrm{Rot}(-2\phi)\sigma_z.$$

The frame loop has lifted angle $\pi + 2\phi$ and winding +2. The component loop has angle $\pi - 2\phi$ and winding -2. The general component law is $\sigma_z = g_{zw}\sigma_w$. Solving for σ_w uses $g_{wz} = g_{zw}^{-1}$. There is one transition integer once the chart order is fixed. Here $(z, w) = (N, S)$, so the tangent transition has winding +2 and the Hopf transition has winding -1. The Bloch map is $(2x, 2y, 1 - |z|^2)/(1 + |z|^2)$. For the round metric, differentiating it gives

$$g = \frac{4(\mathrm{d}x^2 + \mathrm{d}y^2)}{(1 + |z|^2)^2}.$$

Normalizing the coordinate frames therefore replaces $-z^2$ by its phase $-z^2/|z|^2$ on the overlap. Its equatorial winding is still +2. A new unit frame on either cap changes this winding by zero, because its boundary phase extends across the disk and is null-homotopic by Lecture 3. Consequently the oriented orthonormal frame bundle is nontrivial by the cited classification. A nowhere-zero tangent field would give a unit frame and trivialize it, contradicting its winding. This is the hairy ball theorem of Lecture 14, now obtained from transition winding. A unit field and its positive quarter-turn would form a global oriented orthonormal frame. The integer 2 agrees with $\chi(S^2)$ in Lecture 18. Lecture 22 relates this transition winding to the curvature integral.

The angle lifts record the transition in chart order (z, w) and its inverse.

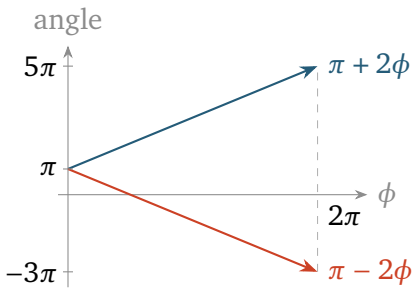


Figure 20.8: The transition g_{zw} has angle lift $\pi + 2\phi$. The inverse map g_{wz} has angle lift $\pi - 2\phi$ and reads components in the reverse direction.

On the equator, use the positive spherical frame (e_θ, e_ϕ), where e_θ points south and e_ϕ points in the direction of increasing ϕ. The normalized coordinate frames have the expressions

$$\widehat{e}_z = \cos\phi\, e_\theta - \sin\phi\, e_\phi, \qquad \widehat{e}_w = -\cos\phi\, e_\theta - \sin\phi\, e_\phi.$$

Their angles in this common frame are $-\phi$ and $\pi + \phi$. The relative angle is therefore $\pi + 2\phi$. The unrolled equator shows both varying arrows in the same tangent-plane coordinates.

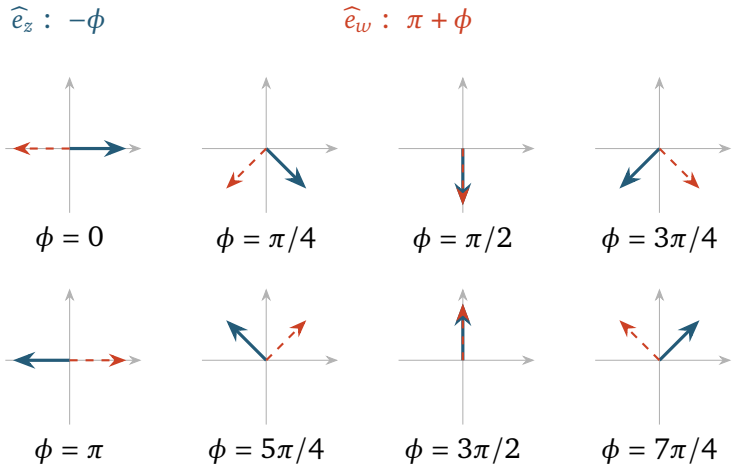


Figure 20.9: At each equatorial point, the horizontal axis is e_θ and the vertical axis is e_ϕ. The normalized z-frame turns through -2π and the w-frame through $+2\pi$, giving relative winding +2.

Self-check. At the equatorial point $z = e^{i\pi/3}$, a tangent vector has complex component $\sigma_z = 2 - i$ in the coordinate frame e_z. Find σ_w and verify that the two frames give the same vector.

Answer. The inverse transition gives

$$\sigma_w = -e^{-2i\pi/3}(2 - i) = 1 + \frac{\sqrt{3}}{2} + i\left(\sqrt{3} - \frac{1}{2}\right).$$

Multiplication by the frame transition gives $(-e^{2i\pi/3})\sigma_w = 2 - i$. Thus $e_w\sigma_w = e_z\sigma_z$. Using the frame transition itself on the component would give the wrong vector.

Summary. Transition functions reconstruct smooth vector and principal bundles. Frames transform by $e_j = e_i g_{ij}$. The same transition gives $\sigma_i = g_{ij}\sigma_j$. A principal bundle is trivial exactly when it has a global section. The Hopf bundle $S^3 \to S^2$ is nontrivial and its associated standard line is tautological. With chart order $(N, S) = (z, w)$, the tangent sphere has transition winding +2. The inverse component map has winding −2. Lecture 21 describes connections and transport in these local frames.

Reading. Nakahara, §9.2–9.4, especially Theorem 9.2. Naber, §§4.2–4.4, especially Theorem 4.3.4, pp. 223–224 (continuous principal reconstruction), and Theorem 4.4.3, pp. 231–232 (topological clutching). Smooth reconstruction follows Nakahara, §9.2.2, with the row-frame convention used here. The homogeneous-bundle proof uses Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.5, for submersion coordinates. The associated-bundle section correspondence also appears in *Lie Groups*, Lecture 26, *From equivariant functions to sections*.